A Paleomagnetic Investigation of Permo-Triassic Red Beds and Andesites from the Spanish Pyrenees

E. J. SCHWARZ

Department of Geological Sciences
University of Toronto, Toronto, Ontario, Canada

Abstract. Fourteen samples of Permo-Triassic andesites and eight samples of Permo-Triassic red beds were collected in the upper Aragón Subordan valley (province of Huesca, Spanish Pyrenees). The measurements were done with an astatic magnetometer, and ac demagnetization was applied to test for stability. The sediments acquired a strong, stable magnetization in post-Alpine times; the original direction of magnetization could not be detected. The magnetic directions of the andesites, on the other hand, probably are caused by thermoremanent magnetism. The pole position is 133°W, 51°N, which differs by about 60° from the mean value of Permian poles determined in northern and central Europe. The best explanation for the difference in the location of the Pyrenean pole and north European pole positions seems to be that there was a northwesterly drift of the western Pyrenees in post-Permian time. Similar results from the Spanish Pyrenees and the Italian Alps suggest that the part of the crust involved in this drifting extended from at least the western Pyrenees to the Italian Alps.

INTRODUCTION

As part of a thesis [*Schwarz*, 1963] a paleomagnetic analysis has been performed on samples of reddish sediments and of andesites which outcrop in the valley of the Rio Aragón Subordan north and east of Oza (province of Huesca, Spanish Pyrenees). The location of the collection site is shown in Figure 1. In this paper the results of the analysis are briefly reported.

SAMPLING AND SOME GEOLOGICAL DATA

Samples were collected from the least weathered rocks. The error in the measurement of the orientation of the samples are estimated to be as much as 3°.

Eight samples (numbers 1–8) were taken from lenses of fine-grained sandstones and from a limestone layer (sample 5). The fine-grained sandstones are locally intercalated in reddish shales. The shale formation, which is roughly 30 meters thick, forms the lower part of a red sedimentary series. The latter may reach a total thickness of 1900 meters. The available evidence does not allow a more precise dating of this series than Permo-Triassic age. Sample 5 was taken from a limestone layer near the top of the series.

Andesite intrusions (see below) are found in the shale formation. Apart from sample 2, which was taken at a distance of 10 cm from the andesites, all samples were collected at least 10 meters from the contact with the andesites.

Fourteen samples (numbers 9–22) of andesites (Johannsen's classification formula is 2212) were taken from sills and a small plug of andesite. The andesite plug is somewhat circular in plan and is visibly connected with the largest sill. The sills occur conformably in the shaly lower part of the Permo-Triassic series. The largest sill is 30 meters thick. In this sill slight compositional differences and an internal discontinuity suggest multiple intrusion. Samples 13, 14, 15, and 16 were collected from the lower layer of the sill.

The following facts indicate that the andesites are of Permo-Triassic age:

1. Similar andesites are found as lava flows within the Permo-Triassic sediments just northeast of the upper Aragón Subordan valley [*Mirouse*, 1959]. This is one of the few areas in the Pyrenees where andesites are found. In each of these areas andesite extrusions are known to occur only in Permo-Triassic sediments; intrusions of andesite are found only in Permo-Triassic and older sediments. Moreover, andesite pebbles may be found in conglomerates, forming part of the Permo-Triassic sedimentary sequence.

2. *Van der Lingen* [1960] investigated comparable andesites and Permo-Triassic sediments 15 km east of the upper Aragón Subordan valley. He reports the 'cleaned' mean magnetic direc-

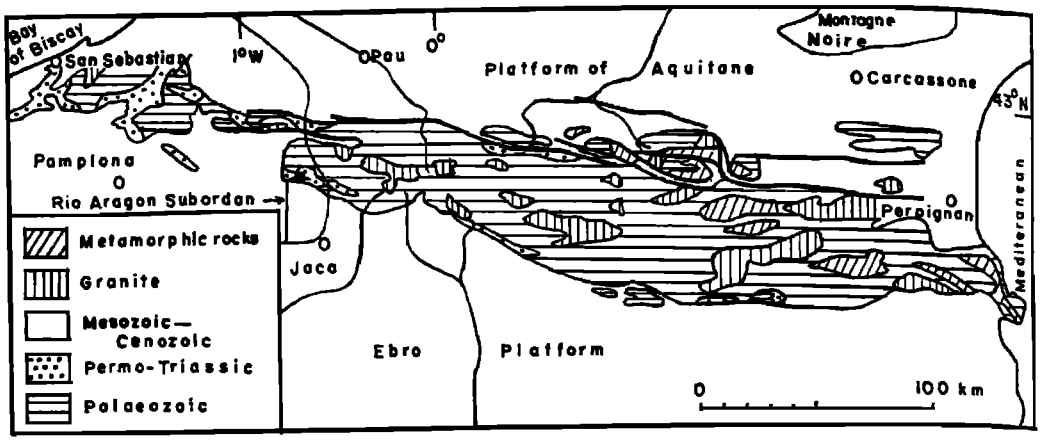


Fig. 1. Geological map of the Pyrenees. Black square indicates the location of the collection site.

tions of andesites and sediments to differ by only 2.5° (α_{63} andesites = 21°; α_{95} sediments = 15.5°). This suggests that the andesites and sediments are penecontemporaneous.

3. A comparison of the orientation of joints in the andesites with those in the Permo-Triassic sediments proves that the andesites have been deformed by the Alpine orogeny [*Schwarz*, 1963]. Consequently, the andesites are older than the Alpine orogenic period (main phase: Eocene).

Method of Measurement

The samples were measured by means of an astatic magnetometer in the Geophysical Department of the Royal Netherlands Meteorological Institute (K.N.M.I.) at de Bilt, Holland. Details on the apparatus and the method of measurement are given by *As* [1960].

The directions of magnetization are represented in a stereographic projection (Figure 2). Table 1 gives the mean magnetic directions, R, and statistical parameters. The declination is taken from north clockwise. The inclination is considered positive if the magnetic vector points downward. The intensity of magnetization of each sample is given in Table 2.

Results of the Measurements

Andesites. All andesite samples show only a minor rotation of the magnetic vector after demagnetization in alternating fields of 900 oe (compare Figures 2 and 3). This indicates that the unstable component has only a small influence on the total magnetic vector. The magnetization of the andesites may be regarded as stable. The mean magnetic direction, R, and statistical parameters are given in Table 1. The magnetic intensities are given in Table 2.

After demagnetization a better concentration of the magnetic vectors was found (see samples possessing similar tectonic corrections in Figures 2, 3, and 4). Moreover, during 'cleaning,'

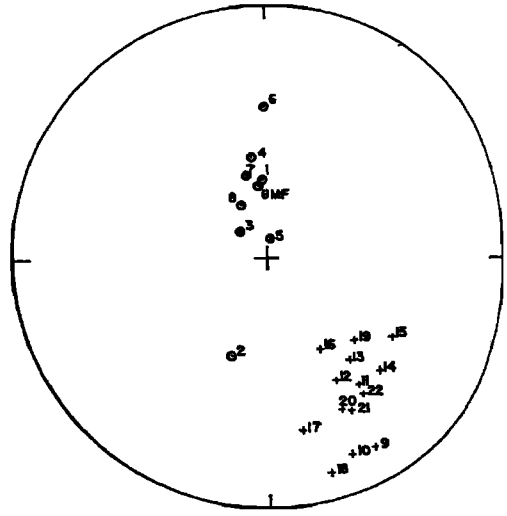


Fig. 2. Directions of magnetization before demagnetization and tectonic corrections. Circles indicate a positive inclination (lower hemisphere). Crosses indicate a negative inclination (upper hemisphere). GMF indicates the direction of the present-day geomagnetic field. Sediment samples: 1–8; andesite samples: 9–22.

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TABLE 1. Mean Magnetic Directions, R, and Statistical Parameters of the Sediment and Andesite Samples before and after Various Corrections

	Sediments				Andesites				
	D	I	R	k	D	I	R	k	α_{95}
Before demagnetization	241	+57	7.0	7	149	−25	13.5	26	7
Without sample 2	248	+58	6.7	19					
After tect. corrections	245	+53	7.1	8					
Without sample 2	256	+55	6.2	8					
After demagnetization	216	+53	6.7	6	149	−23	13.7	49	5
After demagnetization and tectonic corrections	250	+51	6.9	7	152	−22.5	13.8	65	4

D and I, declination and inclination of the mean magnetic direction; R, length of resultant vector; k, dispersion factor; α_{95}, Semiangle of the cone of 95 per cent confidence.

the magnetic directions of most of the samples moved slightly away from the direction of the local, present-day geomagnetic field (indicated by GMF on the diagrams). On the other hand, the samples with significantly larger inclinations moved toward the GMF direction. This caused the inclination of the mean to be smaller after demagnetization than before (Table 1).

Geological phenomena suggest that in the thickest sill intrusion of the andesites took place in two stages. The magnetic directions of samples 13, 14, 15, and 16, taken from the lower layer of the sill, deviate slightly from the rest of the samples (Figure 3). This deviation was probably caused by a difference in age. Sample 19, taken at the base of the plug, shows a magnetic direction which is comparable to those of samples 13, 14, 15, and 16. For this reason sample 19 may well belong to the period of the lower intrusion. In this plug, however, no distinction can be made between two separate intrusive phases. These results suggest that, where geological evidence is lacking, a paleomagnetic analysis may be used to detect different periods of intrusion in igneous rocks.

TABLE 2. Intensities of Magnetization (in 10^{-5} emu per cm^3) before and after Demagnetization

Samples	Before Demagnetization	After Demagnetization
1	1.9	1.6
2	3.9	3.2
3	1.6	1.2
4	1.0	0.9
5	2.7	1.7
6	2.1	1.6
7	1.2	1.0
8	2.6	1.8
9	178	167
10	281	240
11	169	164
12	276	240
13	241	230
14	269	234
15	198	198
16	182	159
17	241	205
18	190	130
19	192	190
20	185	146
21	242	241
22	245	202

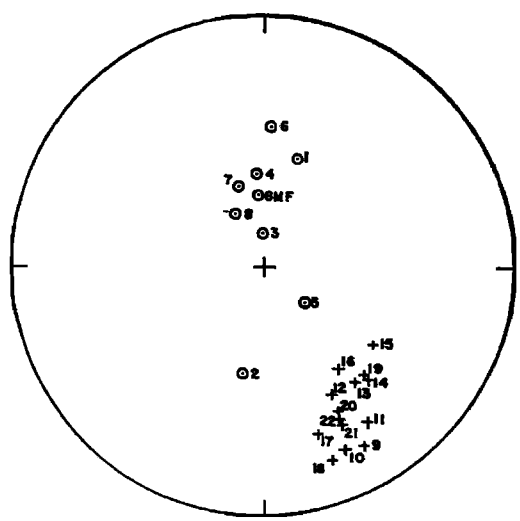


Fig. 3. Directions of magnetization of the sediment and andesite samples, after demagnetization in fields of 900 oe and before applying tectonic corrections.

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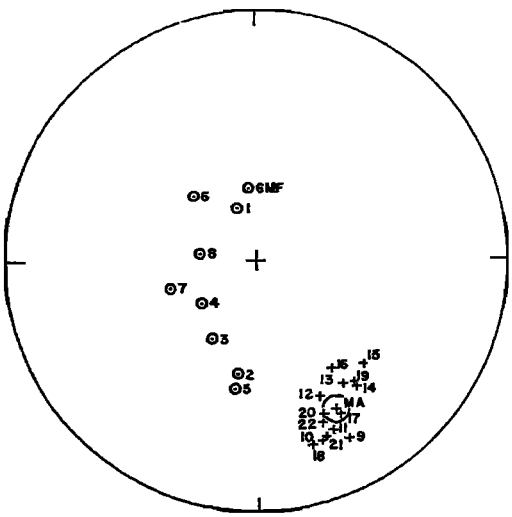


Fig. 4. Diagram showing the directions of magnetizations of all samples after demagnetization and applying tectonic corrections. MA indicates the mean of the magnetic directions of all andesite samples.

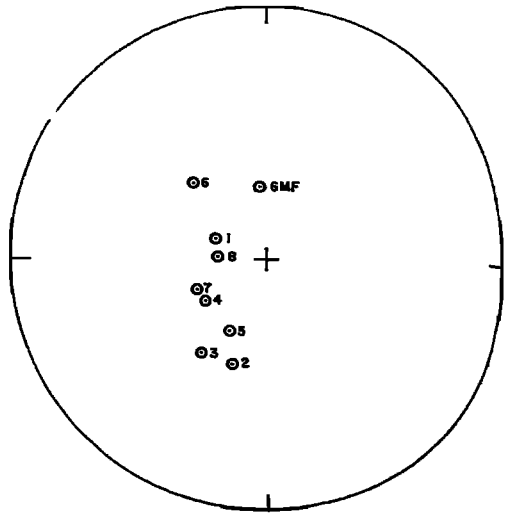


Fig. 5. Directions of magnetization of the sediment samples before demagnetization and after applying tectonic corrections.

Sediments. The directions of magnetization of the sediment samples differ considerably from those of the andesites. Although demagnetization was obtained by applying alternating fields with strengths of 900 oe, the directions of magnetization of the sediment samples show only a minor rotation. Only sample 5, taken from a limestone layer, markedly changed its position (Figures 2 and 3). During demagnetization most of the samples moved away from the direction of the local, present-day geomagnetic field. The magnetic directions for the sediments are closer to the great circle between the GMF direction and the mean direction of the andesites after demagnetization than before. Both before and after demagnetization the point for sample 2 is approximately on this great circle. This is consistent with the geological arguments that the andesites and the sediments do not differ greatly in age. The results are summarized in Table 1.

Red sediments of the type studied usually possess reliable magnetic directions. The magnetic directions of the red sediments in this case, however, are not considered to represent the original ones. In the first place, with the exception of sample 2, the magnetic directions for the sediment samples are in the same general direction as that of the GMF (Figure 3).[1] A concentration of magnetic vectors is apparent although the vectors have not been corrected for tectonic dip, which varies for the samples. Second, the concentration of magnetic vectors is less clear after applying tectonic corrections (Figure 5, Table 1). It is probable that this indicates a recent magnetization which originated after the sediments had obtained their present-day positions during the Alpine orogenic period.

About 15 km east of the Aragón Subordan, on the other hand, *van der Lingen* [1960], measuring sediments approximately similar in lithology and only slightly younger, found a difference of only 2.5° between the mean magnetic directions of andesites and sediments. This indicates that in this area no important later, stable magetization originated in the sediments and that the directions of magnetization reported here approach the original ones.

In the sediment samples the present stable directions of magnetization may be regarded as the resultant of an original magnetic direction and another one which was superimposed on the original. The original magnetization possibly was

[1] Sample 2, which clearly falls outside this group, was taken at a distance of only 10 cm from the contact with the andesites, and it may well have been influenced magnetically by the intrusion.

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acquired at the time of deposition of the sediments, whereas a stronger stable magnetization was acquired later, at least after the time of the Alpine orogeny. Presumably, this later magnetization is due to chemical changes and/or redeposition of ferrous minerals, which form skins around the quartz grains. Such chemical magnetization may have been facilitated by the permeability of the fine-grained sandstones from which the samples were taken.

Pole Determination

Because the sediment samples show a stable direction of magnetization acquired in post-Alpine times, the magnetic directions of the sediments were not used for the calculation of the position of the Permo-Triassic geocentric axial dipole field. The mean direction of magnetization of the andesites corresponds to a magnetic north pole at 133°W, 51°N. The axes of the 95 per cent oval of confidence are 3° and 5°.

Pole Position

From Figure 6 it is evident that the pole position of the andesites differs by about 60° from the mean of Permian poles hitherto determined in northern and central Europe. Similar deviations from the north European poles have been found in another area in the Spanish Pyrenees and in the Italian Alps. Several factors which might be responsible for the deviation between the pole position of the andesites and the north European Permian poles are:

1. *The age of the andesites.* There is sufficient geological evidence to prove that the andesites are of Permo-Triassic age. In the comparison of the pole position of these andesites with those derived from true Permian rocks a difference in age must be taken into account. The deviation of the pole of the andesites from the north European Permian poles, however, is probably less than its deviation from the north European Triassic poles. The age factor may therefore be ignored in the explanation of the anomalous pole position of the andesites with respect to the north European Permian poles.

2. *The magnetic stability of the andesites.* The investigation of magnetic properties of the andesites by means of demagnetization in fields with strengths up to 900 oe revealed good stability. In 'cleaning' most of the vectors moved slightly away from the local direction of the present-day geomagnetic field, and their concentration became denser (Figures 2, 3, and Table 1). The magnetization of the 'cleaned' andesite samples is probably of the TRM type, and therefore may be regarded as the original magnetization.

3. *The original orientation of the plane of intrusion of the andesites.* In the calculation of the pole it was assumed that the andesites intruded along a horizontal plane. If it is assumed that an originally tilted plane of intrusion caused the deviation mentioned above, the orientation of this plane may be reconstructed [*Schwarz,*

TABLE 3. The Locations of the Sampling Sites Belonging to the Pole Positions Given in Figure 6

Locality	Age*	Author
England		
1. Exeter volc.	?	*Creer et al.* [1957]
2. Mauchline sed.	?	*Du Bois* [1957]
3. Mauchline volc.	?	*Du Bois* [1957]
France		
4. Esterel, pyromer. R4	*L*?	*Roche* [1957]
5. Esterel, rhyolite	L?	*Rutten et al.* [1957]
6. Esterel, rhyolite	L?	*As and Zijderveld* [1958]
7. Nideck porphyry	M?	*Nairn* [1957]
8. Montcenis sed.	M	*Nairn* [1957]
Norway		
9. Oslo volc.	L?	*Van Everdingen* [1960]
Germany		
10. St. Wendel sed.	L	*Nairn* [1957]
11. Nahe volc.	L	*Schmucker* [1959]
12. Nahe volc.	M	*Nijenhuis* [1960]
Russia		
13. Ufimskij and Kazanskij sed.	U	*Khramov* [1958]
14. Tartarskij sed.	U	*Khramov* [1958]
Italy (Alps)		
15. Merano porphyries	L	*Dietzel* [1960]
16. Bolzano porphyries	L	*Van Hilten* [1962]
Spain (Pyrenees)		
17. Anayet andesites	?	*Van der Lingen* [1960]
18. Aragón Subordan andesites	?	*Schwarz* [1963]

* L, M, and U indicate, respectively, Lower, Middle, and Upper Permian. The question mark indicates undifferentiated.

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1963]. The minimum dip of the intrusion plane appears to be 29°. It is apparent that such a large dip is unacceptable on geological grounds.

4. *Major tectonic movements during the Alpine orogenic period.* No correction could be applied for tectonic displacements, such as sliding of the Permo-Triassic series with its intruded andesites over the Hercynian core. Sliding may have been accompanied by rotation. It is obvious that a rotation might have affected the original direction of magnetization. The axis of rotation, which agrees with the field observations, is the normal to the intrusion plane. Rotation around this axis, however, affects only the declination of the mean direction; its inclination remains the same.

5. *The nature of the Permian field in Europe.* The paleomagnetic investigations of Permian rocks from northern and central Europe provided data which largely confirm the assumption of a geocentric axial dipole field [*Irving*, 1959; *Cox and Doell*, 1960]. Compared with these data, the measurements in the Pyrenees and the Alps show abrupt changes in declination and inclination.

6. *Continental drift.* The term 'continental drift' is used in order to distinguish clearly between the displacements discussed below and the major Alpine displacements discussed under 4.

It has been seen that the local factors listed above and the nature of the Permian field cannot be held responsible for the anomalous pole position. Of other factors—magnetic anisotropy and magnetostriction, for example—no conclusive data are available. Continental drift is an attractive hypothesis because it offers an explanation of the apparently systematic deviation of the Permian (Permo-Triassic) poles determined in the Pyrenees and the Alps from the poles for northern and central Europe (Figure 6). This discrepancy exists in spite of the fact that samples were collected from various kinds of rocks both north of the Pyrenees and the Alps and in these mountain chains themselves. Moreover, most of the samples were 'cleaned,' and post-Permian magnetization was eliminated. This indicates that magnetic anisotropy apparently has not played an important part in the discrepancy. The direction of the main Alpine stress in the Pyrenees was different from that of the sampled areas in the Alps. Moreover, *van Hilten* [1962] considers the thick, rigid porphyry plate of Bolzano to be only slightly affected by Alpine stress. For these reasons magnetostriction seems to be inadequate as an explanation of the discrepancy.

Van Hilten [1962] has shown that the deviating pole position of the Permian Bolzano porphyries may be explained by supposing a post-Permian drift of the region to the north. Most Alpine geologists favor such a direction of movement. The magnitude of the drift is, however, much larger than is generally accepted. According to van Hilten, only *Carey's* [1958] estimate, based on megatectonic and paleogeographic considerations, corresponds roughly to the paleomagnetic directions for samples from Bolzano. It is thought, however, that although the mean inclination of the Aragón Subordan andesites is 8.5° less than that of the Bolzano

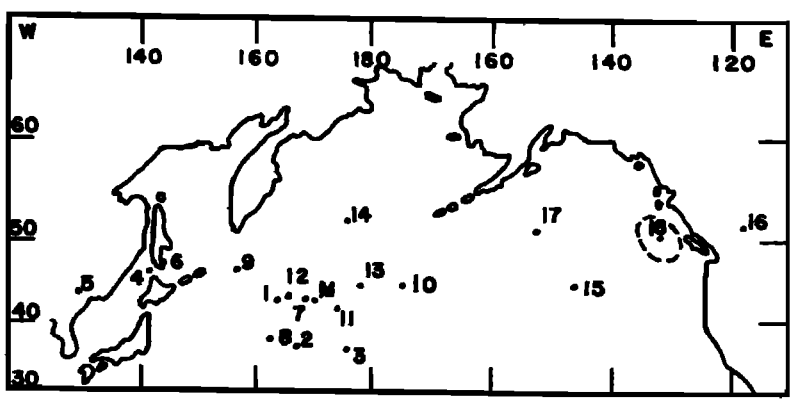


Fig. 6. Permian (Permo-Triassic) pole positions from European sampling sites. *M* indicates the mean of pole positions from northern and central Europe. Numbers refer to Table 3. Revised after *van Hilten* [1960].

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porphyries, the drift proposed by van Hilten would have been much larger because (1) the drift must have been accompanied by a rotation of at least 150° and (2) because of the large extension of the region western Pyrenees–Italian Alps. This implies that the direction of movement would have been to the northwest [*Schwarz*, 1963]. Such a direction of movement is not contradicted by the tectonic and paleomagnetic analysis of *De Boer* [1963] in the Vicentinian Alps (north Italy). More data may be expected from paleomagnetic work which is being done in the Spanish meseta, where structural deformity is small.

Acknowledgments. I am grateful to Professors W. H. Gross, M. G. Rutten, and J. Veldkamp for their stimulating interest during the preparation of this paper. Thanks are also due Mr. J. A. As, who gave me the opportunity to study thoroughly the construction of instruments for paleomagnetic research as used at the K.N.M.I., de Bilt, Holland.

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(Manuscript received December 23, 1963; revised March 4, 1963.)

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